PROJECT PROPOSAL

Milo

Upgrading an open-source quadruped robot into a free-roaming, seeing, hearing, remembering companion — with its body and memory on the robot, and its intelligence on detachable desktop “brains”.

AUTHOR	DATE	STATUS	ESTIMATED BUDGET
Daham Dissanayake	July 5, 2026	Approved baseline	≈ LKR 54,600

Based on the Sesame quadruped by Dorian Todd (open source, Apache 2.0). Full technical detail lives in ARCHITECTURE.md; this proposal summarizes scope, approach, budget, schedule and risk.

01

Overview & Objectives

Milo takes the existing Sesame quadruped — the same 3D-printed body and 8-servo leg layout — and replaces its remote-controlled ESP32 heart with a Raspberry Pi Zero 2W, cameras, microphones, a speaker and an inertial sensor. The result is a robot that walks on a learned gait, sees and recognizes people, listens and talks, and remembers who it has met across power cycles.

The guiding design principle: **body and memory live on the robot; heavy compute lives on the brains.** Any desktop or laptop on the local network running the Milo Brain app can serve as the robot's mind. Milo pairs once via a PIN shown on its face display, then streams camera and microphone data up and receives speech, expressions and movement commands back. Brains are stateless and interchangeable; Milo's identity — its knowledge graph of people, faces, events and facts — never leaves the robot. If no brain is reachable, Milo simply goes to sleep.

	SESAME (BASELINE)	MILO (PROPOSED)
Controller	ESP32 (Arduino C++)	Raspberry Pi Zero 2W (Python 3.11)
Locomotion	Scripted keyframe gaits	RL-trained neural gait (ONNX @ 50 Hz) + CPG fallback
Senses	None	Camera, stereo microphones, IMU
Voice	None	Speaker, TTS, conversational LLM
Memory	None	On-robot SQLite knowledge graph
Intelligence	Remote HTTP commands	Detachable Milo Brain desktop app (any LAN GPU machine)
Face display	OLED, 37 bitmap faces	Same display and art, driven from Python

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System Architecture

The system splits into three independently installable packages sharing one protocol library, so every part is testable off-hardware.

PACKAGE	RUNS ON	RESPONSIBILITY
milo-bridge	Raspberry Pi (robot)	Hardware drivers, 50 Hz gait engine, knowledge graph store, brain discovery, pairing/auth, sleep mode. Runs as a systemd service.
milo-brain	Any LAN desktop/laptop	Voice activity detection, sound-direction, speech recognition (Whisper), face recognition (InsightFace), LLM cognition loop (Ollama), TTS (Piper), system-tray UI.
milo-common	Both sides	WebSocket message framing and PIN-pairing / HMAC challenge–response authentication — byte-for-byte identical on robot and brain.

PACKAGE	RUNS ON	RESPONSIBILITY
training	GPU box (offline)	MuJoCo physics model, PPO reinforcement learning with domain randomization, export of the walking policy to ONNX for the Pi.

All robot↔brain traffic travels on **one multiplexed, authenticated WebSocket**: video (MJPEG, 640×480 at 15 fps) and stereo audio (16 kHz, 20 ms chunks) flow up; TTS audio, face expressions, movement intents and knowledge-graph operations flow down. Brains advertise themselves over mDNS; the robot ranks paired brains by availability and latency and fails over within 10 seconds if one drops.

The cognition loop on the brain: detect speech → estimate speaker bearing so Milo turns to face them → transcribe → match the face against Milo's memory → pull relevant context from the graph → the LLM produces a reply, an expression, a movement and any new facts → speech is streamed back while the talk-face animates, and the facts are written into the robot's graph. Meet someone new and Milo asks their name, then remembers their face.

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Hardware & Budget

The 3D-printed body, eight MG90 metal-gear servos, OLED face and switch carry over from the Sesame build unchanged. New electronics center on the Pi Zero 2W with an 8 MP camera, two MEMS microphones for stereo hearing and sound direction, a 3 W speaker amplifier, an IMU for balance, and a dual-rail battery system — a dedicated 5 A rail for the servos so they can never brown-out the computer.

ITEM	LKR
8× MG90 metal-gear servos	4,640
Raspberry Pi Zero 2 W	23,400
Pi Camera V2.1 (IMX219, 8 MP) + CSI ribbon	9,900
3D-printed body	5,900
Power system (2× 18650, BMS, 2 buck converters, charger)	4,000
microSD card (≥16 GB)	2,500
PCA9685 servo driver board	1,000
2× INMP441 I2S microphones	900
OLED display (0.96″, 128×64)	680
MAX98357A amplifier + speaker	700
MPU6050 IMU	450
Switch, wire kit, heat-shrink	540
Total (approx.)	**54,600**

Compute for the brain side is **zero additional cost**: the Milo Brain app runs on existing machines, with a small tier (6 GB-class laptop GPU) and a large tier (desktop GPU) selecting model sizes automatically.

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Build Plan

PHASE	SCOPE	EXIT CRITERION
0	Parts, SD flash, repository scaffold	Pi boots headless on WiFi
A	Power rails, wiring, hardware bring-up	All buses verified; audio, camera and 8-servo sweep working
B	Bridge core: drivers, poses, faces	Stands, rests, waves and shows faces on battery power
C	Protocol, pairing, streaming, brain skeleton	Live A/V on the brain; pairing under 2 min; failover and sleep/wake
D	Gait training and deployment (CPG first)	Walks and turns on the real floor on velocity commands
E	Hearing and vision pipelines	Turns toward a voice; recognizes a known face; live transcripts
F	Memory graph, LLM loop, speech	Greets a returning person by name; recalls facts across brains and power cycles
G	Robustness, endurance, documentation	All success criteria pass; over 3 hours of battery life

Software-first: every line of Phase B–F code is written and unit-tested off-hardware (mocked buses and models) before the physical build completes, so hardware phases reduce to integration checklists.

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Key Risks & Mitigations

RISK	MITIGATION
Sim-to-real gap in the learned gait	Measure real servo response into the simulator; aggressive domain randomization; a hand-tuned walking fallback exists before RL ships
Pi Zero 2W CPU saturation	Streaming must leave ≥40% CPU headroom before gait work begins; degrade video to 10 fps first
Limited GPU memory on the small brain tier	Tiered model configuration keeps models small; face recognition falls back to CPU
2.4 GHz WiFi jitter	Short 20 ms audio frames with a jitter buffer; UDP escape hatch if needed
Servo current spikes browning out the computer	Dedicated 5 A servo power rail; staggered servo activation; servos never powered from the Pi

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What Success Looks Like

Milo walks and turns on command using its learned gait. It hears a voice, turns to face the speaker, and holds a spoken conversation. It recognizes people it has met and greets them by name. It learns new facts in conversation and still knows them after a power cycle — and after switching to a completely different brain machine. When no brain is around, it curls up and sleeps; a loud sound wakes it to look for one. All of this on a single battery charge lasting more than three hours.

Prepared by Daham Dissanayake · July 2026 · Derived from the approved Project Milo design spec and ARCHITECTURE.md. Sesame quadruped by Dorian Todd, Apache 2.0.